



THE UNIVERSITY OF
NOTRE DAME
A U S T R A L I A

MED 4000

WEEK 28

TUTOR GUIDE

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RELEVANT PRIOR LEARNING:

MED 1000:

- 1.06 Anatomy and physiology of skin [L]
- 1.06 History and examination of skin [L & CCS]
- 1.28 Anatomy of the bones and joints of the upper limb [L & P]
- 1.29 Applied anatomy of the nerves of the upper limb [L & P]

MED 2000

- 2.02 Applied skin anatomy [L]
- 2.02 History and examination of skin & skin procedures [CCS]
- 2.03 Skin cancers [L]
- 2.03 Molecular biology of UV radiation [L]

WEEK 28_SURGERY_PLASTIC SURGERY		
Yr	Wk	LEARNING OBJECTIVES:
4	28	• Know the 3 main anatomical structures which could be affected in patients with acute hand injuries. • Outline the assessment and main findings of patients with nerve, tendon or bone injuries of the hand. • Know which procedures should NOT be carried out in a patient with a bleeding hand. • Outline the management of hand infection.
4	28	• Know the features of carpal tunnel syndrome [CTS] on clinical examination. • List the conditions associated with CTS due to either increase in contents or causing a decrease in the size of the carpal tunnel. • Describe Dupuytren's contracture and the factors known to be associated with this condition.
4	28	• Know the clinical features of seborrhoeic keratosis and outline the management. • Briefly describe the following benign skin lesions: dermatofibroma, skin tags, porokeratosis. • Describe the skin conditions which are precursors of squamous cell carcinoma.
4	28	• Outline the different treatment options for basal cell carcinoma (BCC) based on whether they are high-risk/aggressive subtypes versus lower risk/less aggressive subtypes. • Know the natural history and progression of basal cell carcinoma [BCC] • Know the 'red flag' features in a clinical history or clinical features which would alert the clinician to potential serious problems.
4	28	• Know the indications for specialist referral in patients with suspected melanoma. • Discuss fully the key areas which are important in a medico-legal context in patients with suspected melanoma. • Know the major risk factors for melanoma. • Demonstrate an understanding of a melanoma patient's perspective by contrasting how patients detect melanomas versus the way doctors do. • List the six clinical categories of melanoma.

WEEK 28_SURGERY_PLASTIC SURGERY		
Yr	Wk	LEARNING OBJECTIVES:
4	28	• Outline the principles of wound repair which would minimise the risk of scarring. • Know the difference between split thickness and full thickness skin grafts and the indications for their use respectively. • Briefly describe the 4 types of skin flaps used in reconstructive surgery.

CASE ONE

Short case number: 4_28_01

Category: Musculoskeletal System & Skin

Discipline: Surgery

Setting: Emergency department

Topic: Acute hand injury

CASE

Mark, a 19 year old mechanic, was involved in a fight on a Saturday night. He was unclear about details but he remembered the assailant attacking him with a knife, after which he passed out. His friends brought in to the Emergency Department with his left hand wrapped up in a t-shirt.



Examination confirmed a laceration across the base of his left index finger extending into the palm. He had decreased sensation over half of the index and half of the middle finger. On testing, it was noted that Mark could make a fist. Pain, deformity and swelling of the left wrist was also noted.

QUESTIONS

1. When assessing acute injuries of the hand, what 3 main anatomical structures could be affected? What injury could explain the pattern of numbness noted? Since he could still make a fist, what tendon is affected by this injury?
2. What procedures should NOT be done in a patient with a bleeding hand?

Ten days later he presented with a sausage-like middle finger swelling, with the same finger held in a partially flexed position. Decreased movement of the right middle finger and exquisite tenderness on passive extension was noted.

3. What do these signs indicate? What is the appropriate management?

Resources:

1. Buntine J and Mathias J. How to treat: Soft tissue injuries of the hand. Australian Doctor, March 2004, p 33-40.

Question 1

With acute injuries of the hand, what 3 main anatomical structures could be affected? What injury could explain the pattern of numbness noted? Since he could still make a fist, what tendon is affected by this injury?

His fractured metacarpal was plated and he also underwent repair of the **digital nerve** and **flexor digitalis superficialis** of the left index finger.

a) Exclude NERVE and/or TENDON damage

First decide whether the laceration is superficial or deep and assess damage to the skin edges and wound surfaces. Then consider the possibility of damage to underlying nerves, blood vessels, tendons, ligaments, bones and, less importantly, muscles. Also assess injuries to nails and parts other than the hand. Measure two-point discrimination at the tips of lacerated digits. Observe the resting posture of the fingers with the wrist flexed and then extended. If the patient can co-operate, proceed to test the continuity of possibly divided tendons by requesting isolated active movements. Then test for abnormal passive joint movement, including gliding movement.

Extensor tendon injuries: The middle slip of the extensor expansion of a finger is most easily divided over the dorsal aspect of the proximal interphalangeal joint. Because it develops slowly, the boutonnière deformity that results from this injury is liable to be missed. Gently retract the edges of a finger laceration to inspect the extensor expansion but remember that, over the proximal part of the dorsum of the hand, the ends of completely cut extensor tendons often retract widely. Significant extensor tendon injuries over the dorsum of the hand are recognised by testing isolated active movements.

Take-home messages

- Leave fingertips exposed to monitor the circulation whenever possible. Considerable pain, persistent swelling and even tissue loss can be caused by tight dressings, and long-term stiffness can result.
- Minor hand injuries can lead to major disabilities.
- All nail-bed injuries (except quite minor injuries) are best referred to minimise permanent nail deformities.
- Fine monofilament sutures should usually be left in the hand and wrist for two weeks so that the fingers can be exercised without disrupting the sutured lacerations.
- Slings are useful for short periods but make sure that the shoulder is put through a full range of movement several times each day to prevent a frozen shoulder.
- Discourage patients from covering hypersensitive hand scars and from taking analgesics for painful scars. Desensitisation is required.
- Early return to work is beneficial, both physically and psychologically.
- MRI and CT scans are best arranged by the treating specialist.
- Infections not subsiding after one course of antibiotics should be referred to a specialist.

Flexor tendon injuries: Inspection is less often helpful, although some partially divided flexor tendons may be recognised by inspection. Test isolated active movement of all possibly injured tendons.

b) Exclude underlying bone damage

If only a limited initial clinical examination is possible, X-rays may be arranged immediately or, alternatively (providing the hand has not been severely injured), a resting splint that holds the wrist and digits in a position of function can be applied, together with a sling. The patient can be seen again a day or two later, at which time a thorough clinical examination can be performed, usually followed by radiological examination.

Question 2

What procedures should NOT be done in a patient with a bleeding hand?

- Don't apply a tourniquet as a first-aid measure except for a short time for very severe bleeding.
- Don't use any tourniquet that can crush tissues or can be accidentally left on, eg, a rolled-up glove finger tourniquet.
- Don't apply artery forceps unless you really know what you're doing because you may further damage an artery or a nerve or injure a nerve that was not previously injured.
- Don't give a large initial dose of an opioid analgesic.
- Don't inject a local anaesthetic before trying to test nerve function, especially sensory function.
- Don't take dog, cat and human bites lightly because severe infection may result. Small wounds may contain tooth debris. Refer wounds resulting from a punch onto another person's upper teeth.
- Don't forget to put on sterile or at least surgically clean gloves.

Question 3

Ten days later he presented with a sausage-like middle finger swelling, with the same finger held in a partially flexed position. Decreased movement of the right middle finger and exquisite tenderness on passive extension was noted. What do these signs indicate? What is the appropriate management?

The signs indicate the presence of an infection. Most hand infections follow a minor or severe injury e.g. puncture of the skin by a thorn or splinter or a superficial or deep laceration. The opportunity to prevent infection after a minor injury is often lost because of a late presentation. The risk of infection of a laceration is greatly reduced by thorough debridement, including the removal of all foreign bodies, often with local anaesthesia and a tourniquet such as a finger tourniquet. An infected or infection-prone wound, e.g. a dog bite, should be left open or the skin edges tacked together with small sutures, leaving gaps between them. For an infection-prone wound, eg, a dirty or neglected wound or a crushing or extensive laceration, at least one antibiotic should be administered. Penicillin alone is a poor choice. For an animal or human bite, metronidazole should be considered to prevent infection with anaerobic organisms. In addition to prescribing antibiotics, remember to immobilise the hand and wrist in a position of function using a plaster of Paris slab and to elevate the arm in a sling. Severe infections require appropriate IV antibiotic therapy and elevation of the immobilised hand.

For clean and minor wounds an antibiotic is unnecessary. However, for deeper wounds or wounds with a suspected underlying fracture, anti-staphylococcal cover with oral dicloxacillin, flucloxacillin or cephalexin is indicated for 5-7 days. An initial IV dose of flucloxacillin or cephalothin can be given before referral.

Tetanus immunisation

If the immune status is incomplete (ie <3 doses) or unknown, tetanus toxoid should be given for all wounds but, if the wound is not clean or is of major extent, immunoglobulin should be considered. For patients who have had adequate immunisation at some stage, no treatment is required if the last dose was within five years, or within 10 years for minor wounds. Tetanus toxoid should be given for at-risk wounds if the last dose was within 5-10 years or for all wounds if the last dose was more than 10 years ago. Note that there is no indication for immunoglobulin in an adequately immunised person.

CASE TWO

Short case number: 4_28_02

Category: Musculoskeletal System & Skin

Discipline: Surgery

Setting: Dermatology outpatients clinic of an urban teaching hospital

Topic: Hand surgery_Carpal tunnel syndrome & Dupuytren's

CASE



Sven Morgensen is a 65 year old retiree who presents with history of a 6 month history of pain in his left hand. This pain often wakes him at night, and is partially relieved by hanging his hand over the edge of the bed. He has also noted some numbness and tingling in part of his hand over the last couple of months. What bothers him most is that lately he has had difficulty picking up objects like paper clips and now has trouble doing up his shirt buttons.



On examination there was wasting of left thenar eminence. Sven's grip on the left hand is noticeably weaker compared to the right. He has decreased sensation (on two point testing with a paper clip) over the left index and middle fingers.

An incidental finding is fibrous contracture on the palmar aspect of the right hand causing the little finger to be partially flexed. There are no other abnormal findings in the right hand.

QUESTIONS

1. What is carpal tunnel syndrome (CTS) and what are the classical signs found on examination?
2. What are the 2 main categories of conditions which predispose patients to developing CTS?
3. What is Dupuytren's contracture (DC)? List the known factors associated with DC.

Resources:

1. Henry MM and Thompson JN. Clinical Surgery, 2nd edition. Chapter 33, Arthritis: Hand and wrist, p 659-660. Saunders Elsevier 2005.
2. Conolly B. Carpal tunnel syndrome. Australian Family Physician, September 2009 Vol 38, No 9:684-6.

Question 1

What is carpal tunnel syndrome (CTS) and what are the classical signs found on examination?

Carpal tunnel syndrome is due to median nerve compression in the limited space of the carpal tunnel. The history is usually typical and diagnostic with the patient waking with nocturnal numbness, tingling and pain in a median nerve distribution. The pain radiates to the forearm. The fingers feel swollen but usually are not. Thickened ligaments, tendon sheaths or bone enlargement can cause it, but it is usually idiopathic.

On examination, wasting of the abductor pollicis brevis occurs if the condition has been longstanding. Sensory loss is usually noted in the palm and radial three and a half fingers. *Tinel's sign* may be positive, i.e. reproduction of the pain on tapping the nerve in the carpal tunnel. *Phalen's test* is pain of the palm on maximal wrist flexion.

Question 2

What are the 2 main categories of conditions which predispose patients to developing CTS?

Median nerve compression may arise either from an increase in the contents of the carpal tunnel or a decrease in the size of the tunnel. CTS may be acute, subacute or chronic.

a) Increase in the contents can be due to conditions such as:

- flexor tenosynovitis and its variants
 - inflammatory
 - traumatic
 - gout and allied conditions
- acute swelling from injury (eg. Colles fracture)
- space occupying lesions (eg. simple tumours, cysts, abnormal tissue such as hypertrophy of musculature)
- bleeding into the canal.

In patients with CTS, the most significant clinical finding is the presence of a flexor tenosynovitis and this may be substantiated by ultrasound findings. The fluctuation in the clinical symptomatology of the condition is probably associated with a varying degree of swelling of the tenosynovium. At operation this clinical and ultrasound evidence of flexor tenosynovial swelling is not always evident.

b) Factors causing a decrease in the size of the canal may be either transient/permanent:

- transient conditions – high range repetitive flexion and extension movements which diminish volume and increase pressure in the canal
- permanent conditions:
 - fracture dislocations of the canal walls
 - congenital narrowing (possibly familial)
 - connective tissue thickening (eg. scarring, flexor retinacular thickening).

Over and above these two mechanisms there are many contributing metabolic and endocrine factors. These include:

- rheumatic disease
- thyrotoxicosis myxoedema
- acromegaly
- hormonal – pregnancy and menopause

- diabetes mellitus
- obesity, particularly in association with diabetes.

Treatment of CTS is with a splint to hold the wrist in dorsiflexion overnight. This relieves the symptoms and is diagnostic; used nightly for several weeks it may produce full recovery. If it does not, a corticosteroid injection into the carpal tunnel helps in about 70% of cases, although it may recur. Persistent symptoms or nerve damage require nerve conduction studies showing prolonged latency across the carpal tunnel and surgical decompression of the carpal tunnel.

Question 3

What is Dupuytren's contracture (DC)? List the known factors associated with DC.

Dupuytren's contracture is a painless, palpable fibrosis of the palmar aponeurosis, with fibroblasts invading the dermis. It causes puckering of the skin and gradual flexion of the affected fingers, usually the ring and little fingers. As the lesion progresses, the fingers gradually develop fixed flexion deformities, and the thumb web may also be involved.

The precise cause is not known, but there are nevertheless a number of well-recognised associated factors, including:

- a family history
- male sex
- regular high consumption of alcohol
- insulin-dependent diabetes mellitus
- adhesive capsulitis
- phenytoin therapy
- trauma.

Contracture is bilateral in 45%; similar lesions in the plantar aponeurosis (Lederhosen's disease) occur in 5%; and the penile fascia (Peyronie's disease) is affected in 3%.

Intralesional steroid injections may help in early disease and some advocate transcutaneous needle aponeurotomy. Plastic surgical release of the contracture is restricted to those with severe deformity of the fingers.

Although many measures have been tried, none is effective. The condition may be self-limiting but is usually progressive. Five operations may be considered, and these are outlined in the table below. The choice depends on the extent of the disease, the degree of disability and whether or not previous operations have been done.

Table 33-8. Operations for Dupuytren's contracture of the palmar fascia

Procedure	Technique	Outcome
Percutaneous fasciotomy	Simple division of fibrous bands through percutaneous stab wounds	Recurrence rates high and amage may occur to nerves and blood vessels
Selective fasciectomy	Thickened areas are excised	Most common operation; successful in limited disease
Complete fasciectomy	Whole aponeurosis excised even if not obviously involved	Potentially curative but high incidence of skin necrosis
Complete fasciectomy and skin graft	Aponeurosis excised together with the skin of the palm followed by split-skin grafting	Curative but requires high skills
Amputation	Removal of a single digit (usually little finger), severely contracted into the palm and interfering with hand function	Can be procedure of choice in special circumstances

CASE THREE

Short case number: 4_28_03

Category: Musculoskeletal System & Skin

Discipline: Surgery

Setting: General practice

Topic: Benign skin lesions

CASE

Les O'Sullivan, the 86 year-old veteran with actinic keratoses (4_21_5), became hypervigilant about skin cancer following the last consultation when SCC was diagnosed in one of his lesions. He is now very worried about the pigmented lesion on his back, which has been there for decades, and insists on being checked all over to make sure there are no more cancerous lesions.



Examination reveals a raised pigmented keratotic lesion on his back in at the level of the waist. There were no palpable lymph nodes.

QUESTIONS

1. You recognise the lesion as a classical seborrhoeic wart/keratosis (SK). What is the typical appearance of SK and what is the management?
2. List and briefly describe the other common benign skin lesions.
3. Keratoacanthomas are among the group of seven skin conditions (Solar keratosis, Bowen's disease, Keratoacanthomas, Actinic cheilitis, Arsenical keratosis, Cutaneous horn, Porokeratosis) which are of the precursors of SCC. Describe the lesion and management.

Resources:

1. Henry MM and Thompson JN. Clinical Surgery, 2nd edition. Chapter 38, Skin disorders: Tumours, p 753-4. Saunders Elsevier 2005.
2. Welsh B. The summer skin check. Australian Family Physician. Jan/Feb 2004, p 37-42.

Question 1

You recognise the lesion as a classical seborrhoeic wart/keratosis (SK). What is the typical appearance of SK and what is the management?

Seborrhoeic keratoses are very common benign tumours which are common in the elderly. These are rough lesions which may catch on clothing; this or other trauma may cause minor bleeding. The appearance is unattractive, and cosmetic distaste is a common reason for presentation.

Raised, well-circumscribed lesions may occur anywhere on the body, although the trunk is the most common site. They are initially flat with varying amounts of pigmentation. The surface is waxy with superficial clefting and fissuring.

They develop gradually as slightly elevated, ovoid yellow-brown, well defined papules with a greasy appearance later thickening to become brownish-black. The surface is irregular and shows minute pits. When flat and pigmented, seborrhoeic keratosis may require biopsy to differentiate them from lentigo maligna melanoma. If there is doubt, excision biopsy is indicated.

If the diagnosis is certain, the growth does not have a dermal component, and therefore excision, with subsequent damage to the dermal collagen and scarring, is generally not necessary unless this is the easiest way to get rid of symptoms. They can be removed with a number of destructive techniques such as cryotherapy, curettage, electrosurgery or shave excision. Shaving back the lesion or curettage and cautery give a satisfactory cosmetic result with the additional benefit of providing a specimen for histological confirmation of the diagnosis.

Question 2

List and briefly describe the other common benign skin lesions.

The other common benign skin lesions are:

- Dermatofibroma which are benign tumours of fibroblasts that are firm, solitary dermal nodules usually found on the limbs of younger people. They have an 'iceberg' effect in that they feel larger than they look. The overlying epidermis is often lightly pigmented and dimples when the nodule is squeezed. No treatment is usually necessary, although biopsy may be performed if the diagnosis is uncertain.
- Porokeratoses are uncommon, distinctive, benign lesions due to clones of epidermal cells that show varying degrees of dysplasia. They develop as annular dry plaques surrounded by a raised fine keratotic wall. Porokeratoses may be isolated or multiple and most often appear on the limbs. None usually needed. If lesions are hyperkeratotic or multiple, cryotherapy may be performed.
- Skin tags lesions are usually found in sites where skin surfaces rub together and the skin is therefore chronically irritated. There is a loose connective tissue core covered by epidermis which is variably pigmented. Skin tags are irritated by clothing and bleed as well as causing local symptoms in areas such as the skin surrounding the anus. Patients often present to request removal. The diagnosis is obvious to the naked eye. Management is by snip excision and cautery, which are carried out under local anaesthetic.

Question 3

Keratoacanthomas are among the group of 7 skin conditions (Solar keratosis, Bowen's disease, keratoacanthomas, actinic cheilitis, arsenical keratosis, cutaneous horn, porokeratosis) which are of the precursors of SCC. Describe the lesion and management.

Keratoacanthoma can be thought of as a well differentiated forms of SCC characterised by spontaneous resolution. They are most common on the chronically exposed sites of the head, neck, hands, and forearms. They rapidly enlarge over a period of 4–8 weeks to form a tender, erythematous nodule with a central keratotic plug. Resolution takes 6–12 weeks, sometimes leaving a scar. A minority will persist as SCC.

This lesion arises from squamous epithelium. It is most common on exposed areas of the body and is thought to result from minor trauma. As its name suggests, it has a central keratin plug with a surrounding collar of acanthotic thickened epidermis

Partial biopsy will almost always be reported as a well differentiated SCC as pathology requires the entire architecture to suggest the possibility of keratoacanthoma. If in doubt, treat as a SCC with one of the following approaches:

- Curettage +/- cryotherapy
- Shave excision
- Surgical excision
- Intralesional methotrexate
- Superficial radiotherapy.

Extra caution is required in immunosuppressed patients as keratoacanthomas may progressively grow rather than involute.

CASE FOUR

Short case number: 4_28_04

Category: Musculoskeletal System & Skin

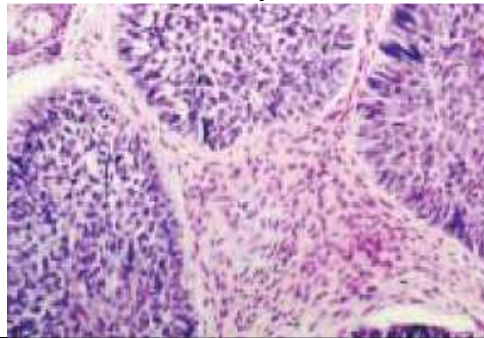
Discipline: Surgery

Setting: Dermatology clinic in a teaching hospital

Topic: Skin cancers_BCC

CASE

Les O'Sullivan (4_21_5, 4_27_3) has been attending the dermatology clinic for decades for treatment of actinic keratosis. One of the AK lesions which underwent malignant transformation was treated with Moh's micrographic surgery. He also has basal cell carcinoma lesions on his face which have been followed up for many years. Imiquimod had been used successfully to treat a few superficial BCCs. On this visit, one particular BCC lesion raised concern because



shiny papular nodules were noted with surface telangiectasia (T).

Punch biopsies were taken and they confirmed nodular BCC.

QUESTIONS

1. What is the risk of this nodular BCC lesion metastasizing? Describe the 'red flags' which would alert you to potential problems with BCCs.
2. **Nodular BCC is the most common type and comprises 60% of BCCs.** What is the clinical significance of the different BCC types? Outline the main treatment options for BCC.
3. What are the 23 Golden Rules in managing skin cancer?

Resources:

1. Hunt MJ. How to treat: Basal cell carcinoma. Australian Doctor. March 2006, p 27-34.
2. Dixon AJ. Managing skin cancer 23 golden rules. Australian Family Physician. August 2005, Vol 34(8):669-71.

ANSWERS

Question 1

What is the risk of this nodular BCC lesion metastasizing? Describe the ‘red flags’ which would alert you to potential problems with BCCs.

Basal cell carcinoma is rarely fatal (metastasising in 0.0028–0.1% of tumours), although it can be locally destructive. Significant morbidity can be associated with the treatment required to eradicate deeply and/or widely infiltrating tumours.

‘Red flags’ which can alert to potential problems include:

- High risk patients:
 - older patients with fair skin, visible sun damage and a history of large amounts of occupational and/or recreational sun exposure
 - organ transplant recipients
 - those with a past history of multiple solar keratoses and/or nonmelanoma skin cancers
- A history of a non-healing or recurring skin ulcer or erosion, particularly if adjacent to an area of previous skin cancer treatment.
- A lesion developing in an area of previous skin cancer treatment, especially if the original lesion was incompletely excised or treated nonsurgically
- A history of new or changing skin lesions.

Question 2

Nodular BCC is the most common type and comprises 60% of BCCs. What is the clinical significance of the different BCC types? Outline the main treatment options for BCC.

BCC subtypes according to histopathological growth pattern and potential for aggression:

- Less aggressive:
 - Superficial
 - **Nodular**
- More aggressive:
 - Infiltrative
 - Micronodular
 - Morphoeic

Table 6: Summary of main treatment options for BCC

Treatment	Indications	Advantages	Disadvantages	Tips
Standard surgical excision	Nodular BCC; some superficial BCC	Histological confirmation of tumour margins; easy wound care; high cure rate; good cosmetic result in experienced hands; surgical scar improves with time	Linear scar; pain; risk of bleeding, infection, wound dehiscence and damage to neurovascular structures; the equipment required	Curette lesion first to define margins; use 3:1 ellipse; hide scar in skin lines
Curettage and electrodesiccation	Superficial BCC on the trunk and limbs	Quick, easy	Slow to heal; white scar; no histological assessment of margins	Stretch the skin; use vigorous curettage, light diathermy
Radiotherapy	When surgery is contraindicated; perineural disease; as palliation in advanced tumours	Painless, good early cosmetic result	Multiple treatments; poor long-term cosmetic results; contraindicated on some body areas (eg, limbs)	Perform biopsy first; curette biopsy useful to de-bulk tumour
Mohs' surgery	High-risk BCC; young patient; cosmetically important site	Histological assessment of all margins; highest cure rates	Special skills and treatment in specialist centre required; time-consuming; expensive if not privately insured	Refer high-risk BCC for Mohs' surgery
Imiquimod cream	Superficial BCC	Excellent cosmetic results	Protracted treatment (six weeks); unacceptably low cure rate for nodular BCC	Stress to patients the importance of breaks in treatment if markedly inflamed (reduces risk of scarring)
Photodynamic therapy	Superficial BCC; nodular BCC <2mm depth	Excellent cosmetic results	Painful, expensive	Minimise pain using local anaesthetic (plain); zimmer cooler, water spray, etc
Cryotherapy	Limited use; small, low-risk superficial BCC on trunk or proximal limbs	Quick, easy	Slow to heal; white scar; poor cure rate	Always biopsy first; mark out BCC and margin before treatment; never use on head or neck BCC

Question 3

What are the 23 Golden Rules in managing skin cancer?

1. Suture rule – if histology indicates further treatment is needed, leave sutures in.
2. Melanoma rule – cut it out early and cut it out widely.
3. Basal cell carcinoma rule – BCCs ain't BCCs.
4. Squamous cell carcinoma rule – SCCs can spread.
5. Dysplasia rule – dysplastic naevi need to be diagnosed (not necessarily excised).
6. Hole rule – think first of the mole, next of the hole.
7. Examination rule – with bright light and magnification, less tumours will be missed.
8. Dermoscopy rule – dermoscopy diagnoses dark dudes.
9. Histology rule – send it to the lab, not the bin.
10. Nerve rule – perineural invasion on histology means radiotherapy consult.
11. Follow up rule – examine rest of skin.
12. Research rule – we are supposed to be scientists.
13. GP rule – GPs rule!
14. Photography rule – two or more: number and photograph them.
15. Computer rule – an Australian GP beats any machine at diagnosing skin cancer
16. Experience rule – the more we see and treat, the better we get
17. Dressing rule – wounds heal better occluded.
18. Time rule – if you only ask one question, make it: 'How long has it been there?'
19. Orientate rule – mark or stitch one edge of tumour for orientation purposes.
20. Pink rule – persisting pink 'pimple' might be nodular melanoma.
21. Efudix rule – GPs pick lesions for Efudix, not patients.
22. Location rule – tumours on eyelids, nose and ear require considerable expertise.
23. Curette rule – careful technique produces best results when curetting SBCCs or Bowen tumours.

CASE FIVE

Short case number: 4_28_05

Category: Musculoskeletal System & Skin

Discipline: Surgery

Setting: Rural general practice

Topic: Melanoma_Principles of skin grafting

CASE

Declan O'Sullivan, the 51 year old son of Les (4_21_5, 4_27_3 and 4_27_4), presents with a lesion on his left cheek because it has enlarged recently. Both he and his sister Sinead had inherited their mother's fair skin and freckles.

On examination this amelanotic papular lesion was noted on the cheek, near the left eye against a background of extensive actinic keratosis. No lymph nodes were palpable.



Because of Declan's family history and skin type, you decide to refer him to the dermatology clinic where Les has been attending. The biopsy taken at the clinic confirmed desmoplastic melanoma. The team consisting of dermatologist, radiation oncologist, plastic surgeon and social worker met to discuss the case.

QUESTIONS

1. What are the indications for referral of suspected melanoma cases? Discuss the key areas to be considered in relation to the medico-legal aspects of such a case.
2. What is the proportion of Australian men and women who are likely to develop melanomas at some time in their life? What are the major risk factors for melanoma?
3. Discuss how patients would detect melanomas and contrast with how doctors detect melanomas. List the six clinical types of melanomas.

Declan is worried about scarring after watching his father's treatments over the years. The general consensus from the case discussion was to offer him surgical excision followed by repair with a flap. He was interested in finding out more about the procedure. The social worker showed him these pictures to give him some idea of how another patient fared after extensive surgery with skin flaps with nose lesions.



QUESTIONS

4. What are the principles of wound repair to minimise risk of scarring?
5. Describe the difference between split thickness and full thickness skin grafts.
6. What are the indications for using flaps in reconstructive surgery? Outline the pros and cons for the latter.

Resources:

1. **Australian Cancer Network Melanoma Guidelines Revision Working Party.** Clinical Practice Guidelines for the Management of Melanoma in Australia and New Zealand. Cancer Council Australia and Australian Cancer Network, Sydney and New Zealand Guidelines Group, Wellington (2008). www.nhmrc.gov.au/files/nhmrc/file/publications/
2. **Pan Y and Kelly J.** How to treat: Melanoma. *Australian Doctor*. November 2009, p 22-28.
3. **Henry MM and Thompson JN.** *Clinical Surgery*, 2nd edition. Chapter 37, Principles of plastic surgery, p 736-40. Saunders Elsevier 2005.
4. **Wu T.** Plastic surgery made easy. *Australian Family Physician*. July 2006, Vol 35(7):493-6.

Question 1

What are the indications for referral of suspected melanoma cases? Discuss the key areas to be considered in relation to the medico-legal aspects of such a case.

Table 2: When to refer melanoma

Difficult histopathology: 10% of patients seen in multidisciplinary melanoma clinics have a significant alteration in diagnosis or micro-staging based on review by pathologists
High-risk primary melanoma
Metastatic melanoma
For evaluation of other pigmented lesions
Consideration of adjuvant protocols or research: currently for stage IV disease there is only limited adjuvant therapy with minimal benefit; a multidisciplinary program fosters collaboration among specialists in clinical and basic research
For specialised melanoma procedures
Anxious patient

Errors related to diagnosis are the most common source of error in general practice, comprising 26–78% of identified errors. Studies reveal a complex mix of up to eight causes per incident, including:

- difficulties in doctor-patient communication
- poor coordination of care between health professionals
- stress in the doctor
- lack of an appropriate management plan
- not accepting limitations in expertise.

Figures obtained from a major medical defence organisation reveal that over a fifteen-year period (1990–2005) there were 47 civil and unlitigated claims. The majority of claims arose from **failure to diagnose melanoma, particularly in cases which were brought to the clinician's attention.**

Table A1 Claims – National Medical Defence Organisation (1990–2005)*

Reason for claim	Surgeons (8) [†]	Dermatologists (2)	GPs (44)
Total number of claims	7	2	38 [‡]
Failure to diagnose patient observed melanoma (delayed or missed)	3	1	32
Incorrect pathology report	2	1	5
Problems in surgical technique and treatment	2		3
Failure to inform patient of diagnosis			2
Failure to obtain pathology report			1
Delay in referral for definitive specialist treatment			1

a) Patients with melanoma will usually present with a history of change in size or colour of the lesion, a change in surface characteristics or elevation of part of the lesion. The history of change in a melanoma is usually measured in months. Alternatively, the pigmented lesion may be noted to look different from other naevi, either by the patient, or by relatives or friend – even though there is no history of change. A melanoma may arise from clear skin as well as from a pre-existing mole.

The key to the clinical diagnosis of a pigmented melanoma is irregularity of the lesion. Irregularity of colour is most important and the presence of a variety of colours in any one lesion is a key feature. Irregularity of outline is the second most common feature with indentations and outgrowths around the lesion often apparent. Irregularity of the surface is another important sign.

b) Psychosocial needs of patients: Many people with melanoma and their carers face practical, emotional and psychological demands in addition to the physical effects of the disease and treatment. Challenges in melanoma include the existential fear faced by anyone with a diagnosis of a life-threatening disease, pain and discomfort associated with treatment, and body image changes associated with disfiguring surgery. Patients with deeply indented scars, such as occur with skin grafting following removal of skin, subcutaneous and deep fascia, as well as those whose scars are longer than they anticipated, may be particularly distressed.

While most studies have found that patients adjust well to melanoma in the long term, distress is common. In a prospective survey of 144 patients with stage I melanoma who did not relapse, 21% reported moderate to high levels of distress three months after excision, 26% at seven-month follow-up and 29% at 13-month follow-up. The impact on families of those with melanoma is also considerable, as they share in the fears of relapse, the traumas of treatment and the sadness of late-stage disease.

Patients' psychosocial needs are significant, and frequently go undetected and unmet. Bonevski et al assessed the perceived needs of a sample of patients attending the Newcastle Melanoma Unit. **Patients reported the most unmet needs in relation to health information, psychological issues and melanoma-specific issues.** The authors recommended that patient needs should be monitored routinely in oncology care so that groups of patients with specific needs could be identified.

c) Information re appropriate action: The guidelines have evaluated evidence in chapters that outline diagnosis, including appropriate investigations and biopsy. There are also chapters on the individual forms of cutaneous melanoma, psychosocial aspects of the disease, and palliative care. Paying attention to the evidence in these chapters when making decisions about the appropriate care program to follow should ensure optimal care and outcome. This approach should assist in meeting the highest level of medico-legal requirements.

d) Should a question arise that may lead to a legal claim, it is well to remember that apart from consideration of guidelines, **appropriate professionalism** reassures patients and should ensure appropriate information for patients through observing the following:

- courtesy
- discussing the program of investigation
- discussing diagnosis and options of care
- clear communication
- compassion
- understanding
- skilful care.

Melanoma 'golden rules':

Rule 1: Evaluate patients' risks for developing future melanomas.

Rule 2: Be aware of atypical presentations of melanomas.

Rule 3: **Listen carefully to the concerns expressed by the patient about changing lesion(s).**

Rule 4: Always examine the scalp, the soles of the feet and between the toes during total-body skin examination.

Rule 5: Ensure appropriate aids are utilised to diagnose melanomas, that is, dermoscopy, total body photography.

Rule 6: The results of partial biopsies of pigmented lesions must be interpreted with care.

Rule 7: The long axis of an elliptical excisional biopsy should follow lines of relaxed skin tension.

Rule 8: It is generally better not to attempt a wide definitive margin at the time of initial excision and to avoid complex closure (ie, graft, flap) for excisional biopsy.

Rule 9: Always consider the need for referral to a multidisciplinary service.

Rule 10: Complete skin exam is an essential part of the assessment of a patient diagnosed with melanoma and should be performed at least annually in follow-up.

Question 2

What is the proportion of Australian men and women who are likely to develop melanomas at some time in their life? What are the major risk factors for melanoma?

About **3.3% of Australian women and 4.0% of men will develop melanoma at some time in their life**, but individual risk varies widely about these averages. The aetiology of melanoma is now understood to involve both environmental and genetic factors and individual risk represents the sum of both types of influences. It is reasonable to posit that successful and timely diagnosis of melanoma will be enhanced if clinicians are aware of high-risk groups in the population, and that people in these groups are aware of their status.

Major risk factors for melanoma are, for the most part, well characterised in population-based studies. They include:

- a) **patient age and sex**
- b) **history of previous melanoma or non-melanoma skin cancer**
- c) **family history of melanoma**
- d) including age of onset and multiplicity of any melanoma cases
- e) number of common melanocytic naevi
- f) number of clinically atypical naevi
- g) **skin and hair pigmentation type** and response to sun exposure
- h) evidence of actinic skin damage.

Present alone or in combination, they can substantially increase risk.

a) **Age** is one of the strongest risk factors for melanoma, as with all cancer e.g. a 70-year-old Australian **man** is eight times more likely to develop melanoma in the next ten years (2.5% risk), and a 70-year-old woman three times more likely (1.0%) than is a 30-year-old man or woman (0.3%).

Ethnic origin is also a strong risk factor e.g. in Los Angeles County, USA, in 1993–1997, non-Hispanic whites were, on average, six times more likely to develop melanoma in a year than Hispanic whites, 19 times more likely than blacks, and 24 times more likely than people of East Asian origin living in the County.

b) **Previous melanoma or other skin cancer** is a powerful predictor of future melanoma, with estimated relative risks ranging above ten. The risk is highest in the first 1-2 years after diagnosis and may be partly accounted for by increased surveillance. A history of non-melanoma skin cancer or premalignant lesions such as actinic keratoses (AK) confers relative risks of around 4-fold in meta-analysis.

Melanoma risk increases with **melanocytic naevus** count, whether counted over the whole body or restricted to one body site, such as the arms. Those with the highest counts (> 100 naevi, whole body) had seven times (6.89; 95% CI 4.63–10.3) the risk of people with the lowest counts (< 15 naevi) in meta-analysis.

c) **Family history:** One first-degree relative with melanoma approximately doubles melanoma risk in international meta-analyses (1.7; 95% CI 1.4–2.1). Data from other common cancers suggest that each close affected relative doubles the risk of that cancer. Relatives affected at an earlier age than average, or who have experienced more than one melanoma, further raise the likelihood that strong familial and genetic risk factors are present.

g) **Skin and hair colour**, skin phototype and freckling: Meta-analyses of studies done mainly in populations of European origin show that light vs. medium/dark skin colour, red–blond versus black hair, and blue vs. dark brown eyes, confer risk increases of about two-fold. Similarly, a person with a Fitzpatrick phototype I (burn easily, never tan) is about twice as susceptible to melanoma as one with phototype IV (always tan, never burn), as are the most heavily freckled versus those without freckles. These phenotypes are not independent of one another and while combinations may further increase risk, they do not multiply it.

The dramatic variation of melanoma incidence across ethnic groups is partly accounted for by these variations in skin type, though there are additional genetic (e.g. MC1R variation) and cultural/behavioural determinants (e.g. clothing and sun-seeking practices).

h) **Sun exposure** is the main driver of melanoma incidence at the population level, with significant contributions made by total lifetime exposure, an intermittent pattern of exposure and exposure in childhood and adolescence. There are no reliable clinical surrogates for prior amount or pattern of sun exposure since they are all confounded by the effect of individual susceptibility to those endpoints. However, the presence of **actinic damage** is detectable clinically and confers a relative risk of 2.0 (95% CI 1.2–3.3).

Question 3

Discuss how patients detect melanomas and contrast with how doctors detect melanomas. List the six clinical types of melanomas.

How patients detect melanoma

About half of melanomas are detected by the patient and present with a history of a new and/or changing lesion. It is important to pay close attention to any history of change, even if the lesion shows no typical clinical features of melanoma. While most are asymptomatic, there may be sensory changes (most commonly itch) that, if persistent, can be helpful in raising suspicion. Changes that are seen in an early melanoma include changes in size, shape or colour in a macular (flat) lesion and, as the melanoma invades more deeply in the dermis, raising up, bleeding or crusting. The time course of the change in melanoma is generally over months. Aggressive, rapidly growing melanomas are more often detected by patients and often present with a nodule, which may be red in colour or pigmented. Patients find changes more useful than atypical clinical features in distinguishing melanoma from benign lesions. All patients seeking advice about pigmented lesions should be encouraged to regularly check their skin with the aid of a mirror or a partner and advised about the changes to look for in early melanoma.

How doctors detect melanoma

The melanomas detected by doctors may be found opportunistically at health examinations for other purposes or at skin examinations for the purpose of detecting skin cancer. Examination for melanoma detection requires examination of the whole skin surface under good lighting. Most melanomas present with an initial flat phase and the features of these melanomas have been summarised by the ABCD(E) rule (Asymmetry, Border irregularity, Colour variation, large Diameter [and Elevation/Evolution]).

Table 1. ABCDE system of diagnosis of melanoma²

A = ASYMMETRY

A lesion is asymmetric if opposite segments of the lesion are appreciably different

B = BORDER

The border of a melanoma is usually irregular, resembling a coastline with bays and promontories around the edge

C = COLOUR

Variation in colour is an important feature

D = DIAMETER

Superficial spreading melanomas are often greater than 6 mm when first diagnosed, but it is possible to diagnose smaller melanomas, particularly nodular lesions which can appear not only as small, shiny dark nodules but also reddish in amelanotic forms

E = ELEVATION

While E designates elevation, it is important to diagnose melanoma while it is flat or with minimal elevation. At this stage the lesion is more likely to be curable

Six clinical melanoma subtypes

The clinical subtypes of melanoma mentioned above were originally described by Dr Wallace Clark and describe presentations of melanoma that have distinctive clinical, epidemiological and body site associations. The clinical aspects are summarised here.

1. Superficial spreading melanoma (SSM) is the most common subtype and is characterised by an initial flat phase that shows changes in size, shape or colour. SSM may occur as early as teenage years and mean age for SSM is in the 40s. Large numbers of melanocytic naevi and more than a few dysplastic naevi are strong risk factors. SSM is associated aetiologically with relatively small amounts of ultraviolet light exposure and has been linked to intermittent exposure and sunburns.

2. Nodular melanoma (NM) accounts for about 15% of melanomas overall, but for the majority of thick melanomas. It presents as a symmetrical, raised, firm, often uniformly coloured and frequently non-pigmented nodule that is enlarging and becoming more raised. Bleeding and crusting are common. NM occurs more often in older people, particularly men, and is more commonly seen on the head and neck than elsewhere.

3. Lentigo maligna (LM) and lentigo maligna melanoma (LMM, the invasive form of LM) accounts for 10–15% of melanomas. It has an initial flat phase that may be prolonged. It presents as an atypical pigmented macule that is changing and has to be differentiated from seborrhoeic keratoses, solar lentigines and pigmented actinic keratoses. LM has been linked epidemiologically to large cumulative doses of UV light, has a strong predilection for the head and neck, and is more

common in outdoor workers, in older people and in association with solar damage and non-melanoma skin cancer.

4. Acral lentiginous melanoma (ALM) accounts for 1–3% of melanomas in Australia and occurs on the acral skin of the palms and soles. It presents with a flat phase with similar appearances and changes to SSM. ALM is more often light-coloured or pink. Importantly, melanomas that appear relatively flat on the soles of the feet may have significant depth histologically. Although the epidemiology is not as well understood, this type of melanoma is at least equally common in people with dark skin and may have no relationship with UV exposure.

5. Subungual melanoma, a variant of ALM, arises within the nail matrix and usually presents initially as longitudinal melanonychia (brown to black stripe throughout the full length of the nail). The differential diagnosis of longitudinal melanonychia includes naevi (moles) or lentigo in the nail matrix, ethnic-type pigmentation (seen with dark skin, often familial and affects multiple digits) and drug-induced pigmentation. The skin of the surrounding nail folds may be involved (Hutchinson's sign).

6. Desmoplastic melanoma may arise within a lentigo maligna or present *de novo*. The latter presentation is typically as a firm, evenly skin-coloured or pink nodule that is progressively enlarging. The differential diagnosis includes dermatofibroma and hypertrophic scar.

Question 4

What are the principles of wound repair to minimise risk of scarring?

Basic principles of wound repair:

- **Atraumatic soft tissue handling:** Tissue ischaemia and desiccation result in necrosis and poor scarring. It is vitally important to avoid rough tissue handling and crushing instruments. Frequent tissue wetting with normal saline prevents tissue desiccation. Judicious cautery can prevent bleeding and improve wound healing.
- **Early repair of traumatic wounds:** Bacterial inoculation of the wound may be increased when the wound repair is significantly delayed. However, when an early repair is not possible, appropriate dressings to reduce tissue desiccation and contamination may allow the surgeon to close the wound later.
- **Thorough wound debridement and lavage:** More reconstructions after trauma fail because of inadequate debridement than because of failed reconstructive technique. Debridement involves the removal of all devitalised and contaminated tissues with maximal preservation of critical anatomical structures.
- **Avoid healing by secondary intention** as it is prolonged and often produces hypertrophic scarring.
- **Judicious antibiotic prophylaxis:** A single prophylactic antibiotic dosage administered just before skin incision may reduce the incidence of wound infection.
- **Appropriate planning of incisions:** In excising a lesion in an elliptical fashion, the long axis should be four times the length of the short axis. When the ellipse is too short, or one side of the ellipse is shorter than the other, 'dog ears' form at the ends of the repaired wound. The final orientation of an elliptical incision should be parallel to the 'relaxed skin tension lines' (RSTL) corresponding to skin wrinkle lines and lie perpendicular to the long axis of underlying muscles. When an excision is planned, it is important to avoid distortion of adjacent anatomical landmarks such as the eyelid, eyebrow or ear helical rim.

- **Expertly executed incisions:** Incisions should be perpendicular to the skin surface and should not be angled. Poorly executed incisions can impair wound edge apposition and result in poor scarring. When tension free direct closure is not possible, reconstruction with either a skin graft or flap will be needed.
- **Layered wound repair:** Deep dermal sutures with buried knots provide the strongest skin wound repair and remove the tension from superficial interrupted sutures.
- **Use appropriate suture size and needle:** By using the smallest possible suture placed correctly, wound repair strength can be maintained and tissue ischaemia can be minimised.
- **Everted wound edges:** Wound edge eversion is achieved by biting a larger amount of the deeper dermis when introducing the needle through the skin.
- **Use of adjuncts such as skin tapes and soft tissue adhesives:** Correctly applied skin tapes such as steri-strips or micropore tape provide wound splinting and reinforce the wound repair.
- **Early suture removal:** When sutures are left in for too long, they become buried and produce cross hatching.

Question 5

Describe the difference between split thickness and full thickness skin grafts.

If a wound is too wide to be closed directly, grafting is the simplest method of repair. A skin graft is a sheet of skin harvested from elsewhere in the body and comprising either its epidermis and superficial dermis (split-thickness graft) or its entire thickness (full-thickness graft). The graft is completely detached from the body and, when applied to the wound, depends entirely on the underlying vascular bed for its nutrition by diffusion of metabolites for a few days before its revascularisation by the ingrowth of blood vessels. A skin graft will therefore only 'take' if applied to wounds which can supply the necessary environment. Graft loss will occur if the wound surface dries out, is infected or contains non-vascular necrotic tissue.

Split-thickness skin grafts are taken with either hand-held graft knives or by powered dermatomes, both of which cut the superficial layers of the skin 150-300 µm thick. As a large number of epithelial remnants are left behind at the donor site, these will, in the same manner as a bad graze, heal within 2 weeks, provided they are kept free from infection.

Large areas of skin can thus be harvested, but as donor sites often heal with some alteration of pigmentation and occasionally with hypertrophic scars, the sites chosen should be inconspicuous, such as the inner thigh or buttock. In the elderly who may have an attenuated dermis, the healing of these donor sites may be considerably delayed. The most acceptable donor site dressing is an alginate mesh, which should be left undisturbed for at least 10 days.

The main disadvantage of split skin is that the final appearance is of a patch of unsightly scarring. Also, the wound continues to contract after graft application, which may lead to disabling contractures across joints and soft-tissue distortion of facial features. Although thin grafts take well, the thinner the graft the more aggressive the contracture.

Given the limited donor sites available for resurfacing extensive defects (like major burns) and the potential morbidity of donor sites, the introduction of the graft mesher proved to be a bonus. The machine cuts slots into the split skin, which enables the graft to be stretched out like a string vest, thereby increasing the area of the original graft. The diamond-shaped slots permit any blood or serum to escape (which might otherwise lift the graft away from its bed) thus reducing the chance of graft failure. Once the graft has taken, each hole heals by secondary intention. However, the disadvantage is that the healed area is like a cobbled surface, implying that it is best suited to inconspicuous recipient beds.

At best, split-thickness grafts must be regarded as 'hole fillers' for the repair of large skin defects.

Full-thickness skin grafts: For smaller wounds in conspicuous areas a full-thickness graft is preferable. Although 'take' of these grafts is less predictable, they retain their original texture with little scarring and contracture, and are particularly suited to the face. However, the colour and texture of the standard postauricular graft is not ideal for all facial areas, and one should consider other donor sites such as

redundant upper-eyelid skin, preauricular and supraclavicular skin to achieve optimum facial skin matching.

Cultured skin: Although not in common usage, there has been considerable clinical experience in the culture of keratinocytes isolated by trypsin from a small split skin graft. These are allowed to proliferate into large sheets of cells on a substrate of fibroblasts. However, such ultrathin layers of cells when applied as a graft were not sufficiently robust to withstand wear and tear and became impractical.

Question 6

What are the indications for using flaps in reconstructive surgery? Outline the pros and cons for the latter.

Simple local flaps can provide better colour, texture and thickness of tissues for the reconstruction. Flaps are particularly useful when a skin graft is not desirable or possible due to poor recipient site factors. The main disadvantages of local flaps include a steep learning curve and additional scarring in the area of the reconstruction from the donor site.

The basic principles of all flap reconstructions are to:

- define the defect first
- find an area where there are redundant tissues and design a way to transfer these tissues to repair the defect which is short of tissues.

The most important part of the flap reconstruction is in its planning. The flap should be designed 'in reverse' starting with defining the defect and transferring it onto a pattern. This pattern is then used to carry out the steps of flap reconstruction in the reverse order until the donor site is reached. The flap pedicle should be of adequate size and not be under excessive tension.

Types of flaps used in reconstructive surgery:

- A **rotation flap** is a semi-circular flap of skin that rotates about a pivot point into the defect. The circumference of the flap may need to be up to six times the length of the defect. The line of the greatest tension is from the pivot point to the distal limit of the defect and can be lessened by a back cut or a Burrow's triangle.
- A **transposition flap** is a rectangular flap of skin that is rotated about a pivot point into an immediately adjacent defect. An example is the 'Limberg flap'. It can be very useful around the temple and the area lateral to the lateral canthus.
- An **advancement flap** is moved directly forward into a defect without any rotation or lateral movement. A back cut or Burrow's triangle may help the flap's advancement. Advancement flaps may be useful around the forehead.
- An **island flap** is based on a vascular pedicle and may allow a greater degree of freedom in the flap's inset. A nasolabial or nasalis myocutaneous island flap may be useful for a lateral nose defect.